

Makhura Kamogelo

Week 3 Practicals

Tasks:

Step 1: Configure DHCP Pools on R1

1. ip dhcp excluded-address 192.168.10.1 192.168.10.1 ip dhcp excluded-address 192.168.20.1 192.168.20.1 ip dhcp excluded-address 192.168.30.1 192.168.30.1
2. ip dhcp pool SALES network 192.168.10.0 255.255.255.0 default-router 192.168.10.1 dns-server 8.8.8.8
3. Repeat for the HR (192.168.20.0) and IT (192.168.30.0) pools.

The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is shown in the 'Physical' tab. It includes a central router (R1) connected to two switches (Switch1 and Switch2). Switch1 is connected to three PCs (PC-SALES, PC-HR, and PC-IT). Switch2 is connected to one PC (PC-IT). The router is labeled 'R1' and the switches are labeled 'Switch1' and 'Switch2'. The PCs are labeled 'PC-PT', 'PC-SALES', 'PC-HR', and 'PC-IT'. The interface shows the 'Logical' and 'Physical' tabs, with the 'Physical' tab selected. The time is 00:58:41. On the right, the 'CLI' window for R1 is open, showing the following configuration:

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0.10
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface g0/0.30
Router(config-subif)#encapsulation dot1q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface g0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
```

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 634, y: 368

Time: 01:08:42

Search for device

Automatically Choose Connection Type

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
!LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
!LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.1
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.1
Router(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.1
Router(config)#
Router(config)#end
Router#
$SYS-S-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool SALES
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.30.1 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
$SYS-S-CONFIG_I: Configured from console by console
Router#
```

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Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 239, y: 20

Time: 01:11:31

Search for device

Automatically Choose Connection Type

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.1
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.1
Router(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.1
Router(config)#
Router(config)#end
Router#
$SYS-S-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool HR
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.30.1 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
$SYS-S-CONFIG_I: Configured from console by console

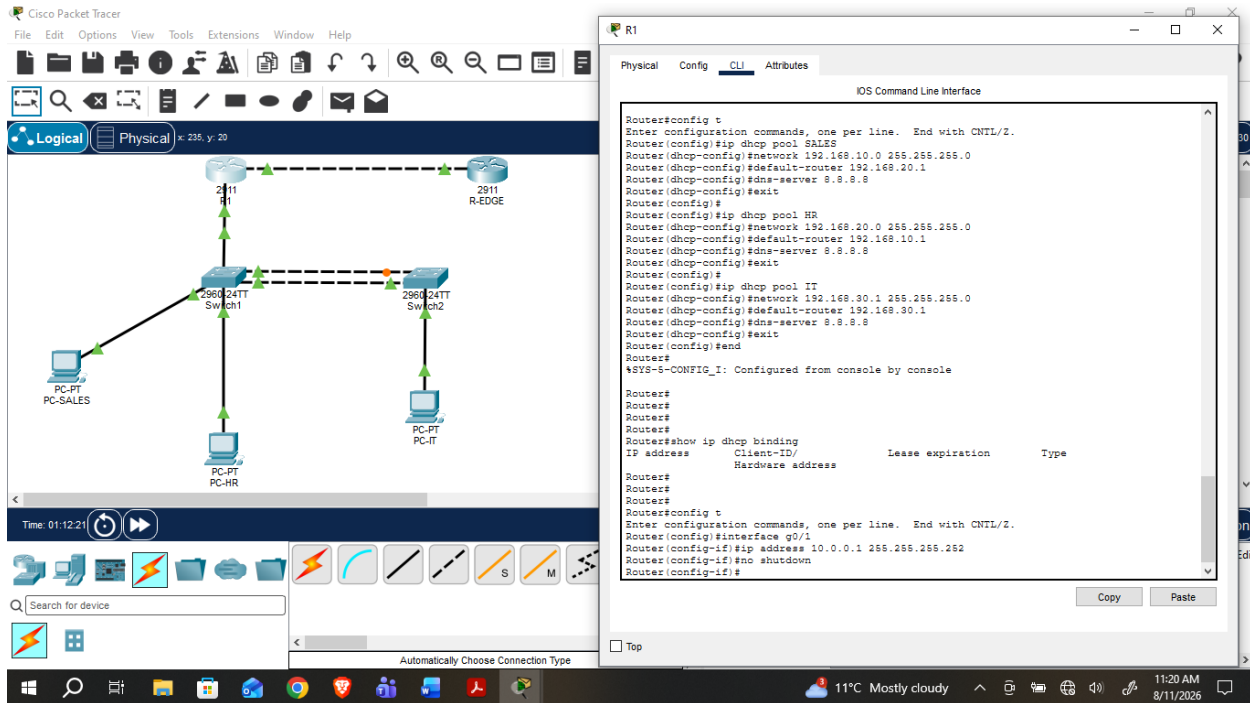
Router#
Router#
Router#
Router#
Router#show ip dhcp binding
IP address      Client-ID/      Lease expiration        Type
                Hardware address
Router#
```

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Step 2: Test DHCP

1. Set each PC's IP configuration to DHCP and verify it receives an address automatically.
2. show ip dhcp binding



Step 3: Static & Default Routing to R-EDGE

1. Assign IP addresses on the R1–R-EDGE link.
2. On R1: ip route [R-EDGE's LAN network] [mask] [next-hop]
3. On R1: ip route 0.0.0.0 0.0.0.0 [R-EDGE's interface IP] (default route)

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 648, y: 78

Time: 01:13:31

Search for device

Automatically Choose Connection Type

R-EDGE

Physical Config CLI Attributes

IOS Command Line Interface

```
Router com0 is now available

Press RETURN to get started.

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
```

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Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 241, y: 22

Time: 01:14:50

Search for device

Automatically Choose Connection Type

R1

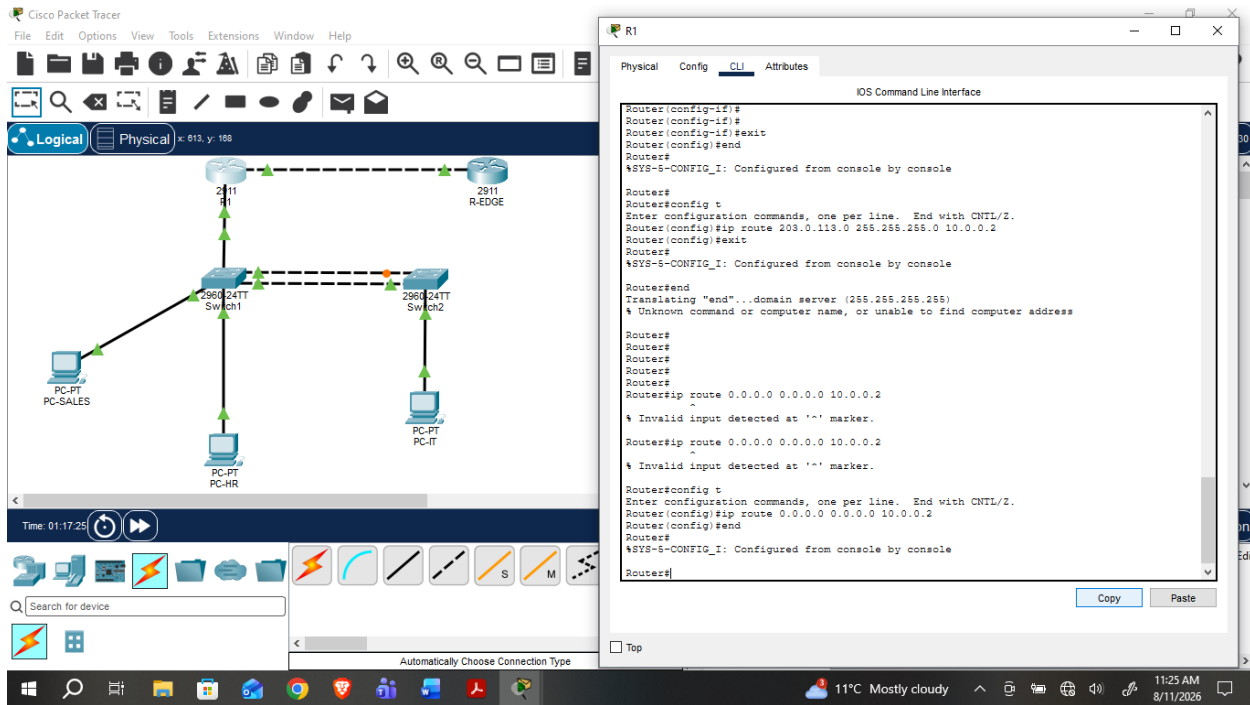
Physical Config CLI Attributes

IOS Command Line Interface

```
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.30.1 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#
Router#show ip dhcp binding
IP address      Client-ID/      Lease expiration        Type
-----
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 209.0.119.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

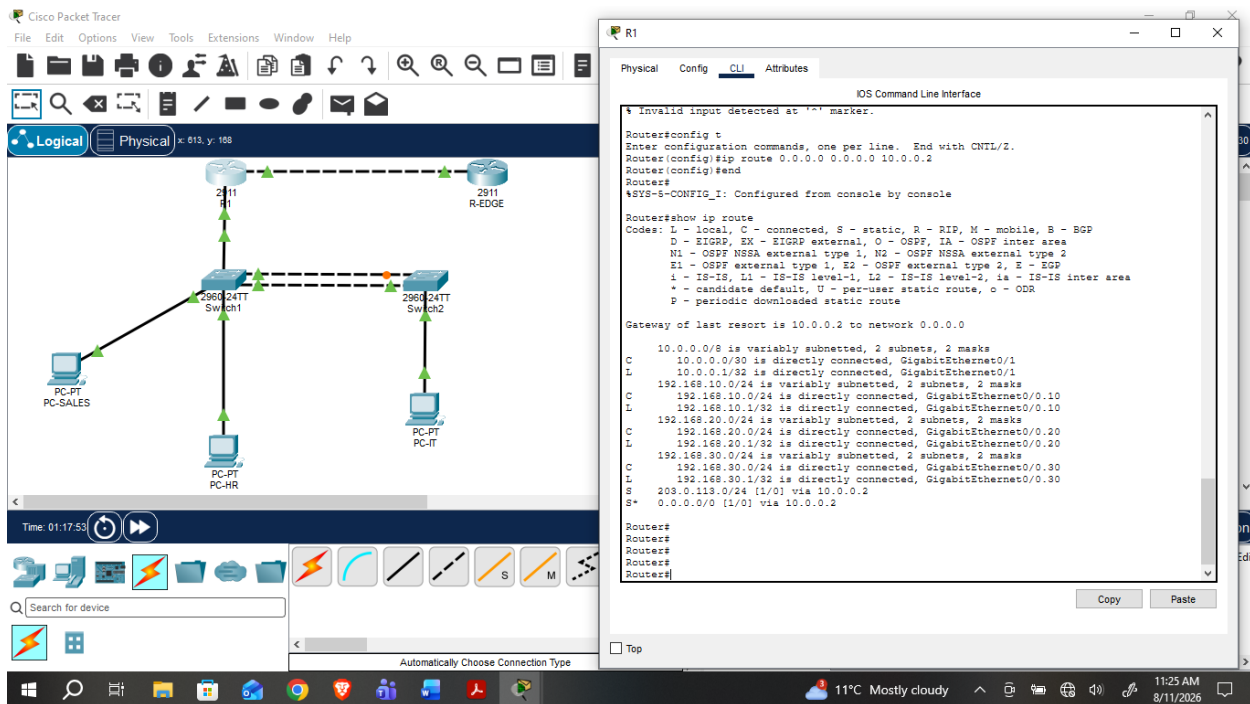
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Step 4: Verify

1. show ip route



Reflection Questions (answer these in your submission):

1. What's the risk of not excluding the gateway address from a DHCP pool?

If you don't exclude 192.168.10.1, 192.168.20.1, 192.168.30.1, DHCP could lease those gateway IPs to a PC. That causes an IP address conflict.

The router and the PC would both claim the same gateway address. Devices in that VLAN would lose their default gateway, so inter-VLAN routing and Internet access would break for everyone in that subnet until the conflict is resolved.

You may see intermittent connectivity and difficult-to-troubleshoot network problems, Devices may lose network or Internet connectivity, and the router/gateway and the client may have the **same IP address**.